How does water reach a leaf?

Learning intention

Explain plant transport and use potometer evidence to infer how air movement affects water uptake.

Mission: trace materials, then test a transport explanation.

A moving bubble is evidence—but evidence of what?

What moves and why?

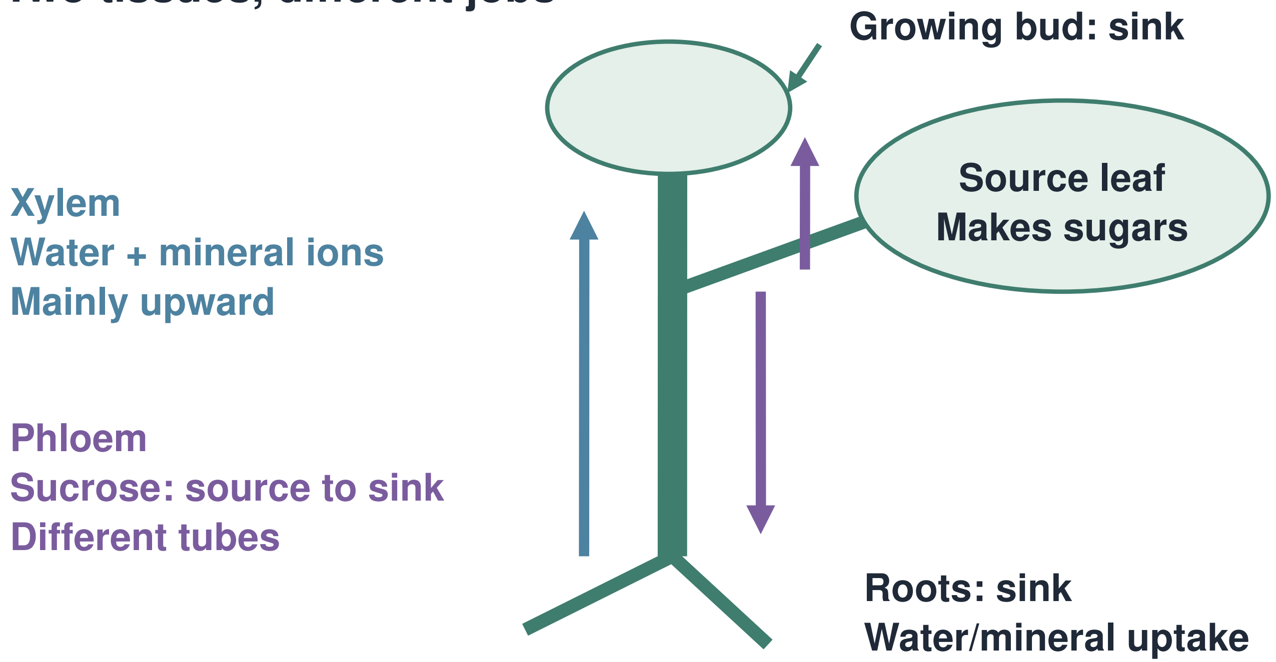
R1. What useful sugar does photosynthesis first make?

R2. Name the process by which water crosses a partially permeable membrane.

R3. How do you calculate a rate?

I follow one material at a time

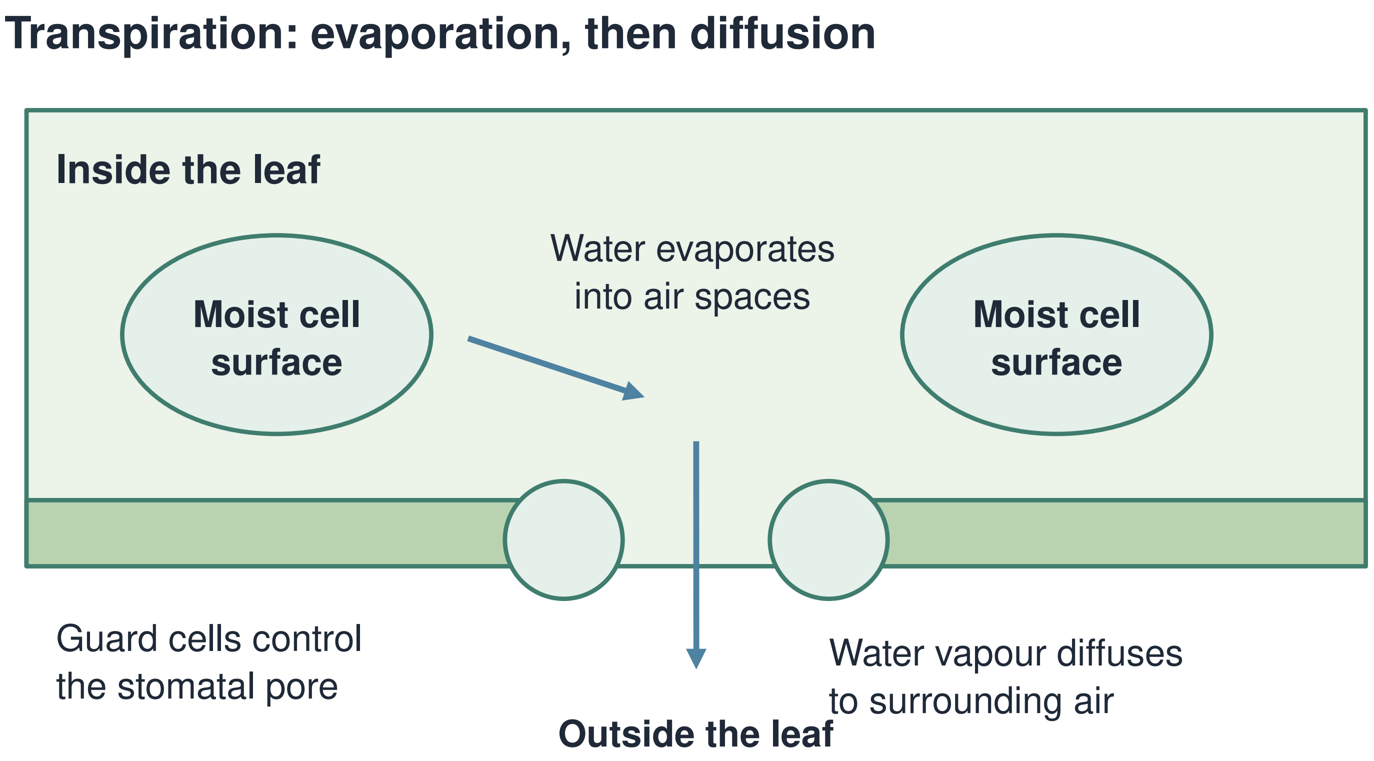
Two tissues, different jobs



Phloem transport uses living cells and energy.

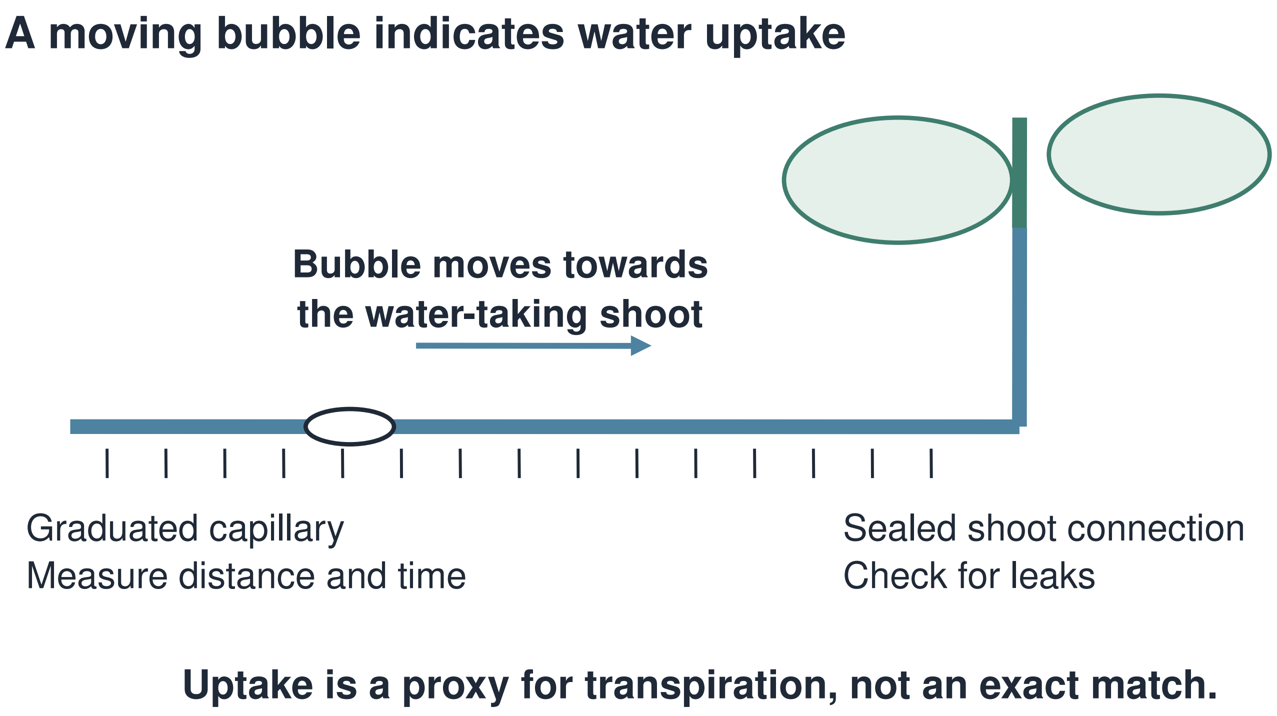
Xylem carries water and mineral ions. Phloem transports dissolved sucrose from source to sink. Different tubes can carry sucrose in different directions.

The leaf links water loss to transport



Water evaporates from internal moist surfaces. Water vapour diffuses through stomata. This loss helps draw water through xylem.

I name the measured quantity



Bubble distance ÷ time gives movement rate. Worked example:15 mm ÷5 min =3 mm/min. Water uptake is a proxy for transpiration.

Air-movement investigation

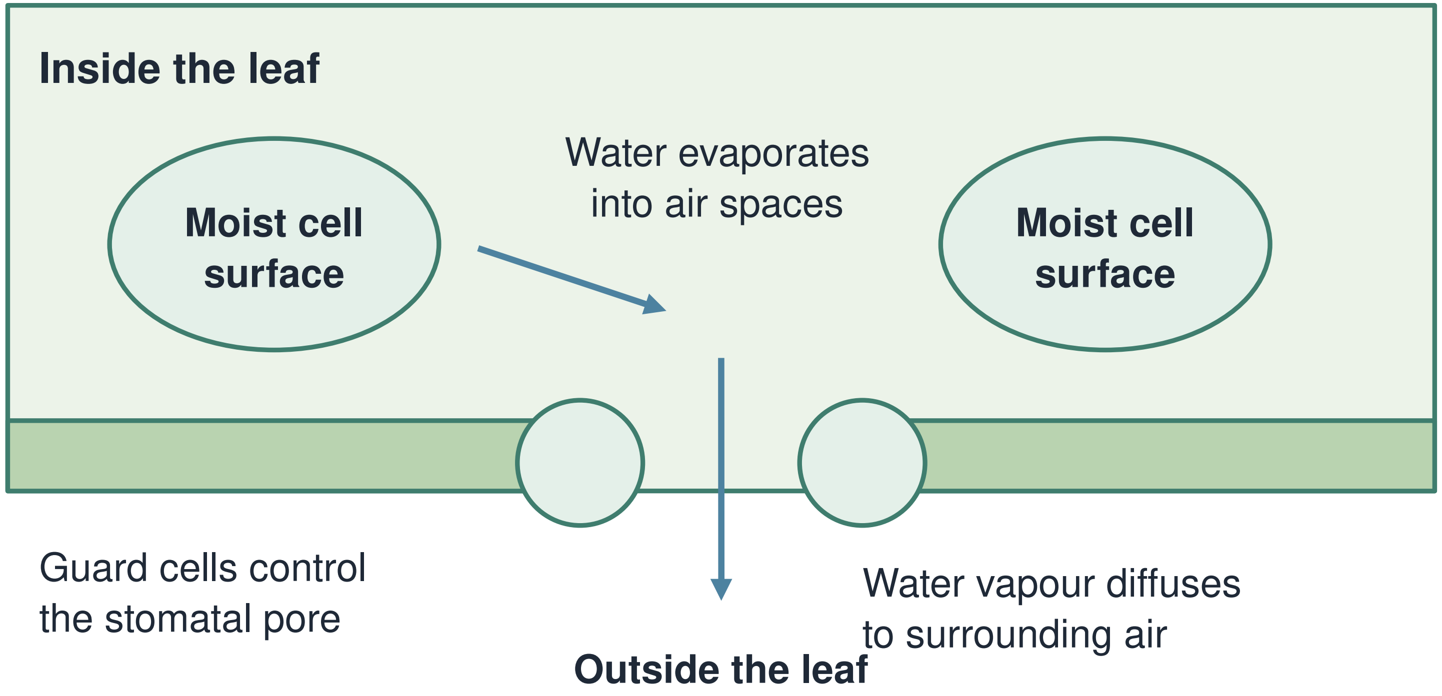
W1. Find the mean movement in each condition. Divide by 5 minutes to compare rates.

Air condition	Bubble movements(mm)	Minutes per trial
Still air	12,11,13	5 each
Moving air	30,28,32	5 each

Constructed repeats; same shoot/capillary/light and 20°C.

Trace the explanation backwards

Transpiration: evaporation, then diffusion



W2. Moving air gave faster uptake here. Explain from surrounding air → leaf loss → xylem uptake.

Repair the instrument claim

W3. “The bubble proves exactly how much water evaporated.”

What should the label say instead?

Choose the accurate transport statement

Read the material and direction carefully.

- **A** Xylem carries sucrose made in the roots.
- **B** Water evaporates inside a leaf and vapour diffuses through stomata.
- **C** A potometer directly measures all water vapour leaving a leaf.
- **D** Phloem transport is always downward.

Prepare the transport report

Choose your route.

Show a material path, a rate and a justified limit.

Make the explanation traceable

Material → tissue → destination.

Movement → calculation → inference → limitation.

Audience: follow my arrows

L1. Show one material route or measurement.

Audience asks: “Is that measured or inferred?”

A careful transport explanation

The tissue explains the path.

The leaf explains the loss.

The apparatus defines the measurement.

Name the path and the proxy

E1. Name the tissue carrying sucrose.

E2. 20 mm in 5 minutes: calculate movement rate.

E3. Why is uptake not exact transpiration?

Choose the next useful step

Tissue roles • leaf mechanism • rate • proxy.